# Sylvester: A Benchmark and Empirical Study of Lifecycle-Aware Entity Resolution on Open Financial Data

Sheng Chang Li
Western University
London, Ontario, Canada
sli3244@uwo.ca

## ABSTRACT

We introduce Sylvester, a public benchmark for *lifecycle-aware* entity resolution on financial entities: matching records across mergers, rebrands, spinoffs, and jurisdiction migrations, where the surface form has deliberately changed. It is built from public GLEIF Level-1/2, SEC EDGAR, and Wikidata, and provides 125,519 labeled pairs over 3.3M entities across three held-out regimes: time, event type, and jurisdiction. At its center is a three-pool protocol that separates methods that learn the benchmark's pool from methods that learn the task, with candidates drawn from the released mined pool, an unrelated miner, and retrieval over the full universe. This matters because we deliberately mine name-similar negatives against name-different positives. Two methods exploit exactly that — inverted name similarity (64–70% Hits@1 on time/event) and a gradient-boosted matcher over name features (0.73–0.84 Hits@1, the best name-input scores we observe) — and then collapse on the other two pools. Among methods that see only names, a 22M-parameter MiniLM cross-encoder is the one that transfers: 0.83–0.87 AUROC across the three regimes and 57–61% Hits@1 on time and event (±2.0–2.3 pp, 95% CI), 24–35 pp ahead of the strongest bi-encoder, and, under a uniform single-pair protocol, ahead of all six zero-shot LLMs on both splits. Input, though, matters as much as architecture. Given the full structured registry record — addresses, previous names, liveness, dates, none of which a name-only method sees — a gradient-boosted matcher transfers across all three pools, beating the name-input cross-encoder on-pool by 18 pp ($p < 10^{-4}$) at negligible cost. Architecture then shapes what survives distribution shift: a cross-encoder over that same record wins the fielded setting by 12 pp retrieval-conditional ($p < 10^{-4}$). Much of the lifecycle signal is in the record, and joint encoding preserves most of it. Even so, the best system tops out at 21–39% Hits@1 end-to-end, because first-stage retrieval (Recall@20 of 33–50%) surfaces only a fraction of the true successors: the bottleneck is finding the candidate, not scoring it.

## KEYWORDS

entity resolution, record linkage, financial entities, lifecycle events, GLEIF, benchmark, cross-encoder

## 1 INTRODUCTION

Twitter Inc. became X Corp. Facebook became Meta. Sprint was absorbed into T-Mobile. Pipelines break silently on transitions like these, a familiar failure across fintech engineering, quant research, and compliance. On the surface such records share almost nothing — different legal names and tickers, sometimes different jurisdictions and legal forms — yet they describe the same continuing economic entity.

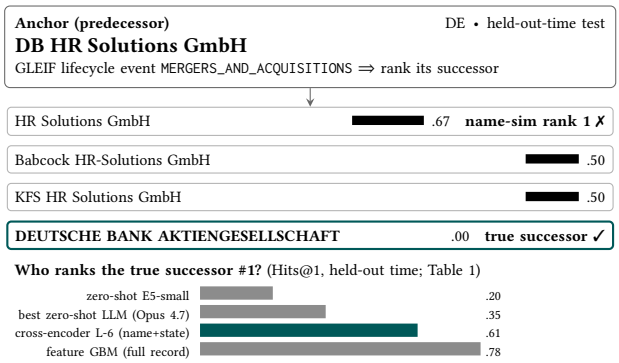



**Figure 1: The task and the headline result. *Top:* a real held-out-time test anchor (4 of 7 candidates; bars = token Jaccard). Methods must rank the true successor above deliberately name-similar negatives; name similarity ranks a coincidental collision first and the true successor — zero shared tokens — last (§6.1). *Bottom:* on this pool the gap closes with *input* more than architecture: zero-shot name scoring fails, a cross-encoder on name+state reaches .61, boosted trees over the full record reach .78 (a name-only exploit hits .77 here but collapses off-pool; Figure 3, §5.4).**

Pair-matching ER treats each record as a snapshot of identity, and does so well: recent LLM-based methods reach 99% F1 on point-in-time records [10]. Adjacent tasks differ. Temporal entity-linking benchmarks like TempEL [12] disambiguate mentions against an evolving knowledge base, and financial knowledge-graph construction [1] works within a single time slice. To our knowledge, no public benchmark targets the question we care about: *given two records observed at different times, do they describe the same economic entity, accounting for any lifecycle event that may have occurred between them?*

One concern is worth addressing up front. The benchmark is engineered against name similarity: negatives are mined to be name-similar, while successors are, by definition, name-different. This is deliberate, documented, and measured, and we report the inverted exploit as its own baseline (§5.4). What the benchmark poses is the discriminative sub-problem that remains once a name-aware retriever has narrowed the field: *given a name-similar candidate pool, what identifies the true lifecycle successor?* (Figure 1.)

*Contributions.*

(1) **An open benchmark** (§3): 125,519 labeled pairs over 3.3M GLEIF Level-1 entities, hard negatives in four classes (the parent/subsidiary class drawn from GLEIF Level-2 relationship records), and three held-out regimes (time, event type, jurisdiction), with construction choices and their scope documented throughout.
(2) **A three-pool diagnostic, and what it separates** (§5): because negatives are mined name-similar while positives are name-different (a property of this benchmark, not of ER in general), inverted name similarity and a name-feature GBM post the best name-input scores we observe on the released pool, then collapse on the other two (§5.4). Among methods that see only names, a 22M-parameter cross-encoder is the one that transfers, beating the best bi-encoder by 24–35 pp Hits@1 and outscoring all six zero-shot LLMs on TIME and EVENT, with gains on M&A and absorption. Give a gradient-boosted matcher the *full* structured record, though, and it transfers to all three too, beating that cross-encoder on-pool by 18 pp ($p < 10^{-4}$); a cross-encoder over the same record then wins the fielded pool by 12 pp. Much of the transferable signal sits in the structured record, where no name-only method can reach it, and joint encoding preserves the most of it.
(3) **The deployment bottleneck is retrieval, not reranking** (§5.6): in a fielded setting over 3.3M entities, end-to-end Hits@1 falls to 15–39%. The rerankers are not the problem — they reach 46–78% conditional Hits@1 — but bi-encoder Recall@20 caps at 33–50%, so the true successor often never reaches them. The system breaks at the first stage, echoing the blocking literature's focus on candidate generation [11].

## 2 RELATED WORK

Classical entity resolution matches records by similarity over structured fields [4], and modern PLM cross-encoders on serialized records (Ditto [8], GraLMatch [3]) are now standard; the OpenSanctions Pairs corpus [10] benchmarks LLM matching at 755K pairs. All of this work adopts a *snapshot* framing rather than lifecycle-spanning matches. Closest in spirit is temporal record linkage [7], which pioneered matching drifting entities but predates modern encoders and open registries. AnyMatch [13] shows that small fine-tuned models rival LLM matchers at a fraction of the cost, and WDC Products [9] studies how hard-negative construction shapes benchmark difficulty, the axis §5.4 examines here. Finally, TempEL [12] time-stratifies entity linking against an evolving Wikipedia, but its task is mention-to-KB disambiguation rather than record-to-record matching over legal entities, and financial-filing knowledge graphs (FinReflectKG [1], from this conference) are built at a single time slice. None of them poses the lifecycle-bridge question directly.

## 3 THE SYLVESTER BENCHMARK

### 3.1 Data sources

We construct Sylvester entirely from open, persistently-licensed sources:

- **GLEIF Level-1 LEI-CDF.** The LEI repository (3.3M entities, 2026-05-19 snapshot), CC0, refreshed every eight hours. Each `LEIRecord` carries legal name, jurisdiction, legal form, addresses, registration metadata, a dated `LegalEntityEvents` log, and a `SuccessorEntity` field linking predecessor to successor by LEI.
- **GLEIF Level-2 Relationship Records.** 473,839 typed inter-LEI relationships, used in hard-negative construction.
- **Wikidata.** High-profile M&A via SPARQL on P156 (*followed by*); a marquee supplement and cross-validation source.
- **SEC EDGAR.** `company_tickers.json` (10,354 records) gives a name↔CIK crosswalk to GLEIF for the U.S. public subset (3,334 confident links); released with the benchmark, not used in the evaluation.

### 3.2 Positive labels

Our gold-standard positives are GLEIF `SuccessorEntity` links: explicit predecessor-LEI → successor-LEI pairs maintained by GLEIF's network of Local Operating Units. The 2026-05-19 snapshot contains 50,202 records with a non-null `SuccessorEntity` element, 33,645 of which resolve to a `SuccessorLEI`. The rest point to successors that never registered an LEI, and we exclude them.

*Label quality validation.* A five-class auto-classifier on the successor pairs (using documented event types, normalized name delta, jurisdiction, legal form) yields: `strong_real` (documented event + name change) 25,428; `likely_real_name_changed` 3,330; `likely_real_event_only` 1,704; `ambiguous` 1,355; **`likely_administrative`** 1,828 (5.4%, filtered out). We retain the first four classes (94.6%; `ambiguous` stays because every link is still LOU-curated), yielding 31,817 plausibly-real predecessors (31,824 positive rows; seven links are duplicated in the source export).

### 3.3 Hard negatives

For each positive predecessor we mine up to 80 hard negatives drawn from four classes:

- `parent_subsidiary` (221 pairs) — LEIs that share a Level-2 relationship-record edge (consolidation, fund-management, sub-fund) with the predecessor. These look related but are legally distinct.
- `same_name_same_country` (10,208) — distinct LEIs with identical normalized legal names in the same jurisdiction: coincidental name collision vs. true lifecycle linkage.
- `same_name_diff_country` (10,703) — distinct LEIs with identical normalized name in a different country.
- `token_overlap_same_country` (232,601) — fuzzy fallback: distinct LEIs in the same country sharing ≥ 2 content tokens after normalization, ranked by Jaccard. Covers the long tail when name-exact collisions are sparse.

After deduplication this leaves 253,733 candidate (predecessor, hard-negative) pairs covering 20,192 of 31,817 predecessors, or roughly 63%. The remaining 37% are small entities in jurisdictions without name collisions, and they fall back to random in-batch negatives during training.

### 3.4 Evaluation splits

For each positive we identify its *anchor lifecycle event* — the highest-priority COMPLETED event on the predecessor's record by type-priority then effective date. We then assign each positive (and its associated negatives) to one of three split regimes:

- **Held-out time:** anchor effective date < 2025-01-01 (train) vs. ≥ 2025-01-01 (test); completed-event volume rises ≈4× in 2022 when GLEIF tightened reporting, so the effective window is 2022–2026. Test: 3,448 positives, 2,007 Hits@1-eligible anchors (§5.2).
- **Held-out event type:** train on M&A, rename, legal-form change; test on spinoff, demerger, breakup, absorption — dominated by *absorption* (2,609 of 2,760 anchors, 94.5%), the rest statistically thin (§5.5).
- **Held-out jurisdiction:** train on US + GB + DE; test on SG + JP + BR. Test composition: 161 anchors, 161 positives, of which 88 are Hits@1-eligible — the smallest split; its headline numbers carry ∼±8.5 pp 95% bootstrap CIs (vs. ∼±2.0–2.3 pp on time/event).

We refer to these regimes as TIME, EVENT, and JURIS. Each positive gets up to 5 hard negatives if its splits are all `train`, or up to 10 if any split places it in `test`, yielding 125,519 final pair rows.

*Wikidata marquee supplement.* Released alongside the benchmark: 302 high-profile M&A pairs from Wikidata P156 both-sides-resolved to GLEIF, 283 not captured by GLEIF `SuccessorEntity` (Sprint → T-Mobile, Wyeth → Pfizer, Interbrew → AB-InBev); not used in the headline evaluation.

## 4 METHOD

We present two methods at different cost-accuracy points: a bi-encoder (§4.1) that embeds each side independently and scores by cosine (cheap), and a cross-encoder (§4.2) that reads both records together, 2× costlier per pair, producing the headline name-input result.

### 4.1 Bi-encoder

We fine-tune `intfloat/e5-small-v2` (33M params, 384-dim) via supervised contrastive learning with state features folded into the text prompt: "`query:` $n_\ell$ `| jurisdiction:` $c_\ell$ `| form:` $f_\ell$". Training uses the 26,297 positives in the strict no-leakage intersection of all three split regimes (one hard neg per anchor), MultipleNegativesRanking loss (InfoNCE; in-batch + explicit hard negs), 3 epochs, batch 32, AdamW lr $2 \times 10^{-5}$, ≈6 min on Apple silicon. Ablation variants (in-batch-only; E5-cosine hard negs plus 1,372 transitive positives) are in Table 1. The bi-encoder uses contrastive loss and the cross-encoder BCE on the logit — different objectives by architectural convention.

### 4.2 Cross-encoder

A bi-encoder pins each record to a fixed point in embedding space, yet the lifecycle event deliberately changes the record's surface form, so the model is asked to keep two records that look nothing alike close in cosine. A cross-encoder sidesteps this: it reads both records jointly and can compare specific tokens between them.

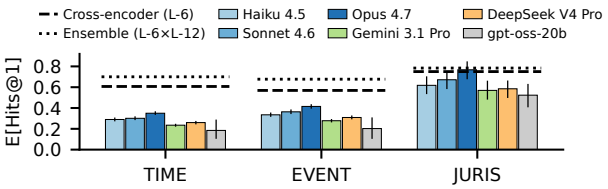



**Figure 2: Zero-shot LLM panel: E[Hits@1] (95% CIs) for six models under the uniform single-pair protocol, against the fine-tuned L-6 (dashed) and ensemble (dotted; Table 1). Every LLM trails the L-6 on the two large splits; on JURIS ($n$=88) Opus meets it. `gpt-oss-20b`: 56/52/50-anchor sample.**

We fine-tune a second-stage cross-encoder that ingests $(\text{record}_A, \text{record}_B)$ as a single concatenated sequence and emits a single relevance logit. The backbone is the 22M-parameter `ms-marco-MiniLM-L-6-v2` cross-encoder from HuggingFace; training uses the same positives as `method_v2_full` (26,297 plus 1,372 transitives) plus a 4:1 sample of E5-cosine hard negatives per positive, with positives labeled 1.0 and negatives 0.0. Loss: standard binary cross-entropy on the logit. 2 epochs, batch 32, lr $2 \times 10^{-5}$, 100 warmup steps. The extensions in §5.3 hold architecture/optimizer constant and modify only the serialized text, the training positives, and the backbone.

*Inference cost.* Scoring all 125,519 pairs takes ≈80 seconds on MPS; in deployment the cross-encoder re-ranks a retriever's top-$K$, and the benchmark's fixed 5–10-candidate pools make direct full scoring feasible (Table 1).

*Hard-negative re-mining.* Re-mining with E5-cosine top-20 same-country candidates, in place of the noisy token-overlap fallback, yields 652,764 negatives across 31,714 predecessors (vs. 253,733 over ≈20K before). The cleaner negatives are what the cross-encoder trains on, to large effect, and they are half of what later unlocks the full-record GBM's transfer (§5.4). For the bi-encoder they change nothing (§5.7).

## 5 EXPERIMENTS

### 5.1 Baselines

We compare against four similarity baselines — `exact_name` (normalized name equality), `token_set` (token Jaccard), `tfidf_char` (char-3gram TF-IDF cosine), and `sentence_encoder` (zero-shot E5-small-v2, the method's backbone un-tuned) — and one *supervised* classical matcher, `features_gbm`: gradient-boosted trees over twelve string-similarity and field-agreement features in the Fellegi–Sunter lineage [4], trained on the released pool's no-leakage rows; the classical name-feature default (§5.4 upgrades it to the full record). Six zero-shot LLMs run under one protocol: one call per pair, byte-identical prompts carrying the same name+state inputs as the neural models (legal name, jurisdiction, legal form), a 0–10 integer rubric, `temperature=0` where exposed. Five API models from three organizations cover the full test set (26,564 pairs each) — **Claude Haiku 4.5**, **Sonnet 4.6**, **Opus 4.7**, **Gemini 3.1 Pro**, **DeepSeek V4 Pro** (thinking disabled) — and **gpt-oss-20b** (a

fourth organization) runs locally on a 56/52/50-anchor sample (Table 1 footnotes).

Paired permutation tests on per-anchor expected Hits@1 separate the comparisons cleanly. The L-6 beats all five API models on TIME and EVENT at $p < 10^{-4}$ (+15 to +37 pp). The 88-anchor JURIS split is closer: the L-6 leads Haiku, Gemini, and DeepSeek ($p \leq 0.019$), sits within noise of Sonnet ($p = 0.14$), and is indistinguishable from Opus 4.7, which holds the higher point estimate (0.767 vs. 0.750, $p = 0.74$). The ensemble beats Opus on TIME/EVENT ($p < 10^{-4}$) and is a coin flip on JURIS. The panel (Figure 2) is strikingly vendor-heterogeneous: every Claude model outscores both other API vendors' frontier models, Haiku included, and §7 examines the elicitation-format confound behind part of that spread. Under Holm correction across the paper's ≈30 paired tests, every $p < 10^{-4}$ result survives; the $p \approx 0.02$ JURIS-level and on-pool-TIME margins do not, and should be read as directional.

## 5.2 Metrics

We report per-anchor Hits@1 and MRR alongside pairwise AUROC, and ship Hits@3, AUPRC, and threshold-tuned F1 with the released metrics. Discrete scorers tie often, so Hits@k and MRR are exact *expectations over uniformly random tie-breaking*; a fixed tie order is biased. Per-anchor metrics cover *eligible* anchors: a pool of at least two candidates with the gold present. The 37% of predecessors with no mined hard negative leave singleton pools and drop out (1,612 of 2,760 EVENT-test anchors are eligible; §3).

## 5.3 Headline results

Table 1 reports test-set metrics across the three split regimes for all baselines and method variants on the released benchmark pool (`pairs.parquet`; token-overlap-mined negatives). JURIS numbers ($n$=88) clear their CI band but stay directional. §5.6 retests on a mechanically different pool, and Table 2 extends the L-6 to the L-6×L-12 ensemble.

*Extensions: enriched prompt + L-6×L-12 ensemble.* Stacking prompt enrichment (GLEIF Level-2 parent/child names, founding year, ISO country, category), 259 Wikidata-mined training positives, multi-seed averaging, and an L-12 ensemble partner lifts Hits@1 +9 to +11 pp on time/event (+3 on jurisdiction, inside its CI; Table 2); baseline comparisons keep the single L-6 apples-to-apples. One negative result: symmetric scoring (averaging $s(a, b)$, $s(b, a)$) hurts −0.9/−1.9/−9.1 pp — consistent with a *directional* predecessor→successor representation (the TIME delta sits within its CI).

## 5.4 The construction cuts both ways: an anti-similarity baseline

`token_set`, `tfidf_char`, and `sentence_encoder` all fall below chance AUROC on time/event (Table 1). The cause is mechanical: hard negatives are deliberately name-similar to the predecessor while positives are by construction name-different, so on this candidate distribution name similarity is *evidence against* a true successor link. So we report the mirror image as its own baseline: `anti-token_set` ranks candidates by *negated* Jaccard, and on the curated pool it is strong (Table 1; *above* the single L-6 cross-encoder

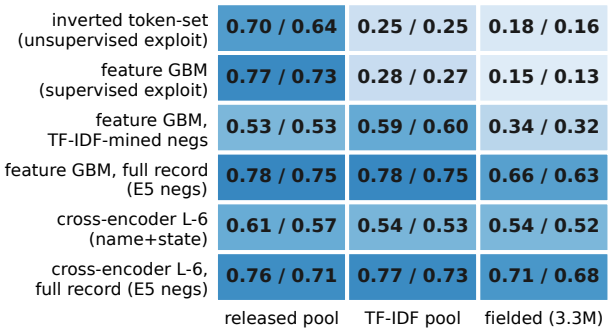



**Figure 3: The argument in one exhibit: expected Hits@1 (TIME / EVENT; fill = mean) as the candidate pool shifts from the released pool to an unrelated miner's pool to retrieval from the full 3.3M-entity universe. Read rows by their *minimum*. The name-only exploits (rows 1–2) collapse off-pool; the name-feature GBM retrained on TF-IDF negatives (row 3) recovers only on its own miner's pool. Among name-input methods only the cross-encoder transfers (row 5). Given clean negatives *and* the full record, trees (row 4) and joint encoding (row 6) both transfer — and the full-record cross-encoder holds the highest minimum and wins the fielded pool (§5.4). The name+state TF-IDF cell re-trains that cross-encoder on E5-cosine negatives (`trn-E5`, §5.6), within 2 pp; fielded cells include the gold (strict conditional and end-to-end in the text); JURIS in the text and tables.**

on time and event) because picking the least name-similar candidate exploits the construction.

The supervised `features_gbm` does the same thing, learned rather than hand-built. Given training labels it discovers the inversion along with the miners' residual signatures, posts the best name-only scores in the paper (Table 1; above the ensemble at $p < 10^{-4}$ on TIME/EVENT), and then collapses off-pool just as hard (Figure 3; on JURIS, 0.36 on Sylvester-TF-IDF and 0.12 fielded; AUROC across splits 0.54–0.57 and 0.45–0.52).

Is that collapse a fact about feature models, or about what we fed this one? It carries two handicaps we can remove, separately and together: the noisy v1 negatives and the name-only feature set. On the cross-encoder's own E5-mined negatives, the name-feature GBM improves cross-pool but still dies fielded (0.25 / 0.25 retrieval-conditional vs. 0.46 / 0.50); a TF-IDF-negative retrain behaves alike (Figure 3, row 3). Given the full structured record (addresses, previous names, liveness, registration status, dates, registrar; the successor fields the labels derive from are excluded) but v1 negatives, it wins the released pool (0.87 / 0.84) yet still fails to cross pools (0.38 / 0.35 on TF-IDF). Given *both*, it transfers across all three and wins (`features_gbm-full`): 0.78 / 0.75 on-pool (+18 pp over the L-6, $p < 10^{-4}$; +8 pp over the ensemble), 0.78 / 0.75 on TF-IDF, and 0.51–0.66 fielded retrieval-conditional, where the enriched ensemble leads by 1–3 pp on TIME/EVENT and trails by

**Table 1: Test-set metrics on the three split regimes, released pool. Bold = headline L-6 cross-encoder used in baseline comparisons; the L-12 matches or beats it throughout, and the L-6×L-12 ensemble (Table 2) is the strongest *name-input* method that survives pool shift (§5.4). Random ranking scores 0.18–0.22 Hits@1 (pools average 7.4–8.3 candidates); the best bi-encoder lifts Hits@1 5–8 pp over zero-shot E5-small and the cross-encoder a further 24–35 pp. Hits@1 95% CIs: ±2.0 / ±2.3 / ±8.5 pp. *Supervised GBMs (§5.4): the name-feature variant and `anti-token_set` (*negated* similarity) exploit the construction — the strongest name-input rows here, 2–4× chance off-pool — while `features_gbm-full`, trained on the cross-encoder's negatives with the full structured record, transfers across all three pools (Figure 3), as does the full-record cross-encoder — the strongest fielded system in the paper (§5.4). Protocol and metrics: §5. †Opus: no temperature param; Gemini (preview): thinking not disableable, lowest level, Batch API. ‡local; 56/52/50-anchor sample.**

| | Held-out time | | | Held-out event | | | Held-out jurisdiction | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | AUROC | Hits@1 | MRR | AUROC | Hits@1 | MRR | AUROC | Hits@1 | MRR |
| `exact_name` | 0.474 | 0.163 | 0.373 | 0.523 | 0.193 | 0.400 | 0.571 | 0.352 | 0.539 |
| `token_set` | 0.193 | 0.136 | 0.319 | 0.266 | 0.180 | 0.356 | 0.323 | 0.356 | 0.522 |
| `tfidf_char` | 0.277 | 0.180 | 0.360 | 0.349 | 0.213 | 0.386 | 0.568 | 0.420 | 0.583 |
| `sentence_encoder` | 0.315 | 0.204 | 0.385 | 0.386 | 0.232 | 0.408 | 0.648 | 0.432 | 0.587 |
| `anti-token_set` (1−Jaccard) | 0.807 | 0.700 | 0.761 | 0.734 | 0.638 | 0.715 | 0.677 | 0.516 | 0.618 |
| `features_gbm` (supervised)* | 0.935 | 0.772 | 0.840 | 0.926 | 0.734 | 0.816 | 0.960 | 0.835 | 0.897 |
| `features_gbm-full` (full record)* | 0.929 | 0.783 | 0.860 | 0.912 | 0.753 | 0.838 | 0.893 | 0.881 | 0.924 |
| *LLM pairwise, zero-shot — uniform single-pair protocol, full test coverage* | | | | | | | | | |
| `Claude Haiku 4.5` | — | 0.290 | 0.476 | — | 0.335 | 0.517 | — | 0.618 | 0.738 |
| `Claude Sonnet 4.6` | — | 0.301 | 0.492 | — | 0.364 | 0.542 | — | 0.672 | 0.758 |
| `Claude Opus 4.7`† | — | 0.350 | 0.534 | — | 0.416 | 0.589 | — | 0.767 | 0.839 |
| `Gemini 3.1 Pro`† | — | 0.235 | 0.433 | — | 0.278 | 0.474 | — | 0.569 | 0.699 |
| `DeepSeek V4 Pro` | — | 0.260 | 0.451 | — | 0.309 | 0.497 | — | 0.585 | 0.712 |
| `gpt-oss-20b`‡ | — | 0.185 | 0.377 | — | 0.203 | 0.403 | — | 0.523 | 0.683 |
| `method_full` (bi-enc, token-overlap negs) | 0.403 | 0.255 | 0.425 | 0.451 | 0.291 | 0.450 | 0.667 | 0.500 | 0.627 |
| `method_no_hard` (bi-enc, no explicit negs) | 0.407 | 0.259 | 0.429 | 0.461 | 0.299 | 0.459 | 0.668 | 0.511 | 0.641 |
| `method_v2_full` (bi-enc, E5-cosine negs) | 0.405 | 0.250 | 0.424 | 0.457 | 0.296 | 0.455 | 0.694 | 0.500 | 0.629 |
| **Cross-encoder (L-6, 22M)** | **0.836** | **0.607** | **0.725** | **0.825** | **0.569** | **0.701** | **0.868** | **0.750** | **0.818** |
| Cross-encoder (L-12, 33M) | 0.844 | 0.631 | 0.742 | 0.836 | 0.598 | 0.724 | 0.870 | 0.750 | 0.823 |
| Cross-encoder (L-6, full record)* | 0.930 | 0.761 | 0.845 | 0.915 | 0.708 | 0.808 | 0.954 | 0.875 | 0.916 |

**Table 2: Cross-encoder extensions: Hits@1 [95% bootstrap CI]; L-6 row repeated from Table 1. Prompt enrichment is the largest increment on TIME/EVENT; the L-12 pairing captures the JURIS gain (CIs overlap — read per-step deltas as directional).**

| Model | TIME | EVENT | JURIS |
|---|---|---|---|
| Cross-encoder (L-6, 22M) | 0.607 [0.588, 0.629] | 0.569 [0.547, 0.593] | 0.750 [0.659, 0.830] |
| + enriched prompt | 0.675 [0.653, 0.697] | 0.642 [0.620, 0.662] | 0.750 [0.659, 0.830] |
| + Wikidata-LEI positives | 0.694 [0.669, 0.712] | 0.651 [0.630, 0.673] | 0.773 [0.682, 0.853] |
| + L-12 backbone (33M) | 0.697 [0.678, 0.716] | 0.676 [0.657, 0.697] | 0.761 [0.670, 0.841] |
| **Ensemble L-6×L-12** | **0.700 [0.681, 0.718]** | **0.677 [0.656, 0.700]** | **0.784 [0.682, 0.864]** |

6 pp on JURIS (all n.s.) — at negligible inference cost. Permutation importance credits fields no name-only method sees: candidate liveness, shared postal/city, previous-name overlap, same registrar. Nor is the transfer a live-vs-dead prior: ablating all four liveness/registration-status features costs 4–7 pp and transfer survives (0.72 / 0.72 on TF-IDF; 0.43 / 0.47 fielded-conditional, at parity with the L-6).

Completing the grid: a cross-encoder trained with the identical protocol but a serialized *full-record* prompt (the same fields the GBM sees) lands 2–5 pp behind the trees on-pool (0.76 / 0.71, $p \leq 0.018$), is within 2 pp on TF-IDF (0.77 / 0.73), and pulls clearly ahead on the fielded pool: 0.64 / 0.66 / 0.78 retrieval-conditional, +12 pp over the GBM and +9–18 pp over the ensemble ($p < 10^{-4}$ on TIME/EVENT); best end-to-end here (21–39%).

The claims accordingly split by input class and by pool. Among name-input methods — every LLM, bi-encoder, string baseline, and name-feature GBM — the L-6 is the only one whose worst pool stays strong. Given the full record, trees and joint encoding tie on curated pools and the cross-encoder wins the fielded universe. Much of the transferable signal is in the structured record: no name-only method, joint attention included, can recover fields it was never given. Joint encoding is the best way we test to hold onto that signal once the distribution shifts.

The unsupervised exploit shows the same pattern at lower amplitude, collapsing under every pool change we test, though on a few event slices it outranks both name-input cross-encoders. On Sylvester-TF-IDF it falls to 0.25 / 0.25 / 0.20 Hits@1 (chance is 0.09) while the cross-encoders hold 0.53–0.66 (Table 4); on the fielded pool it drops to 9–18% (gold-included, as plotted), a fraction of the cross-encoder's (§5.6). Per event it is a one-way bet on acquirer-renaming, and there it is genuinely hard to beat: on the M&A slice of the TIME split ($n$=1,506) it scores 0.741, *above* both

**Table 3: Per-event-type Hits@1 on the held-out-EVENT test split (Hits@1-eligible subset), zero-shot E5-small vs. L-6. The full test split is 94.5% ABSORPTION (2,609 of 2,760); SPINOFF at $n$=4 is statistically uninformative. Lift concentrates in the absorption-like types.**

| Event type | $n$ anchors | zero-shot E5 | Cross-enc (L-6) | Δ (pp) |
|---|---|---|---|---|
| ABSORPTION | 1,546 | 0.230 | **0.571** | +34.1 |
| DEMERGER | 42 | 0.333 | **0.595** | +26.2 |
| BREAKUP | 20 | 0.150 | **0.400** | +25.0 |
| SPINOFF | 4 | 0.500 | 0.250 | −25.0* |

* $n$=4, statistically uninformative.

the L-6 (0.625) and the ensemble (0.711) — the event type the inversion is built for (full-record models still outscore it there). The bet is one-way: the same heuristic scores 0.11 on pure renames and 0.35 on legal-form changes, where the cross-encoder scores 0.90 and 0.85, and it loses pairwise AUROC to the cross-encoder on every split (Table 1). Curated-pool Hits@1 alone can therefore be gamed; within the name-input class, the cross-encoder's claim is being strong *simultaneously* on the curated pool, the alternative-miner pool, the fielded pool, and every event type with a usable sample (Figure 3). A plausible mechanism is joint attention: asking whether the *differences* between two records look like a lifecycle-event pattern, not how similar two strings are.

## 5.5 Per-event-type breakdown

Aggregate Hits@1 hides large variation across event types.

*Held-out time.* Pure renames (CHANGE_LEGAL_NAME, $n$=10) are the easiest slice for every similarity-direction method, zero-shot E5-small included (9 of 10, no lift); only the inverted exploit fails them (0.11, §5.4). The gains come instead from events where the post-event name is inherited from the acquirer and string similarity cannot bridge the gap: MERGERS_AND_ACQUISITIONS ($n$=1,506: 0.18 zero-shot → 0.62 L-6 → 0.71 ensemble; the inverted exploit's best slice, 0.74, §5.4) and ABSORPTION ($n$=461: 0.24 → 0.54 → 0.65). The ensemble also more than doubles zero-shot on the small but operationally critical BREAKUP subset ($n$=7: 0.14 → 0.71).

*"Held-out event type" really means "held-out absorption."* The held-out-event test set is 94.5% ABSORPTION anchors (§3.4); DEMERGER, BREAKUP, and SPINOFF contribute $n$=42, $n$=20, and $n$=4 within the Hits@1-eligible subset. Table 3 shows the lift is consistent across the absorption-like types and that the SPINOFF cell is statistically uninformative, so the aggregate "+34 pp Hits@1 over zero-shot E5" on this split is primarily a transfer claim about absorption, not a strong claim about spinoffs.

## 5.6 Robustness to candidate-pool shift

The headline cross-encoder is trained and, mostly, evaluated against candidates from one family of hard-negative miners: v1 token-overlap and v2 E5-cosine both rank by name-related signals, and the cross-encoder trains on the E5-cosine set. So the obvious question is whether the gap survives an evaluation pool built with a mechanically different miner, or whether the headline number is partly an artifact of how the negatives were chosen.

*Setup.* We re-mine hard negatives with sparse char-3gram TF-IDF cosine, which has no learned representation and shares no parameters with E5, holding the other three classes, splits, and positives constant (Sylvester-TF-IDF). We train cross-encoders on the E5-cosine negatives (trn-E5) and the TF-IDF negatives (trn-TF-IDF) and score each on both pools. The released pool's negatives are token-overlap-mined, so the headline configuration is itself already cross-miner; the only miner-matched cell is trn-TF-IDF on the TF-IDF pool.

*What the 2×2 tells us.* The architecture generalizes across miners: each trained model lands within 1–2 pp of the other on the pool it never saw (TIME/EVENT), miner-matching is worth a bounded +7–11 pp, and the gain over zero-shot E5 survives every pairing (+13 to +40 pp, outside any CI band) — though its size depends on the miner, so read the abstract's "+24–35 pp" in that light.

*Fielded retrieve-then-rerank.* A deployment pipeline retrieves top-K candidates from the full universe, then reranks — unlike the curated pools above. The trained method_v2_full bi-encoder retrieves top-20 per test anchor from the 3.3M-entity GLEIF universe, filtered to the anchor's jurisdiction (the lightest production filter); the cross-encoder reranks. The filter is nearly free (1.8–4.1% of positives cross countries, capping recall at ≥95.9%, far above the observed 33–50%): the bottleneck is the embedding, not the filter.

Bi-encoder Recall@20 is 33.2 / 39.6 / 49.7% on TIME / EVENT / JURIS. That is the hard ceiling for any reranker. When the bi-encoder does retrieve the true successor, the L-6 picks it #1 with 46–57% Hits@1, the ensemble 53–60%, the full-record GBM 51–66% (n.s. vs. the ensemble) — and the full-record cross-encoder 64–78%, ≥ 9 pp clear of every other reranker ($p < 10^{-4}$ on TIME/EVENT; §5.4). End-to-end deployment Hits@1 (recall × conditional, unretrieved truths counted as misses) is 15–28% for the L-6, 17–30% for the ensemble, 17–33% for the full-record GBM, and 21–39% for the full-record cross-encoder. The drop from in-distribution to deployment confirms rather than contradicts §4.2: the bi-encoder's fixed embeddings, which cannot retrieve lifecycle-spanning successors by cosine similarity, are exactly the bottleneck the anti-correlation analysis predicts.

## 5.7 Ablations

*State features help modestly.* Ablating jurisdiction and legal form from the prompt costs 1.1–1.8 pp Hits@1 on every split, most on held-out event type and least on held-out jurisdiction. Scale does not substitute: moving the bi-encoder backbone from E5-small (33M) to E5-base (110M) leaves TIME/EVENT unchanged (−0.2/−1.2 pp; JURIS +2.3), so on this step at least the ceiling looks structural rather than parametric.

*Explicit hard negatives slightly hurt the bi-encoder.* In Table 1, method_no_hard edges out the v1 full model by 0.4–1.1 pp Hits@1 on every split, and E5-cosine re-mining does not change that. The culprit is the token_overlap_same_country class, 92% of mined v1 negatives and the lowest-precision one; in-batch negatives from MultipleNegativesRanking are cleaner on their own. Better explicit negatives, in other words, do not always help contrastive training, even though they are exactly what the cross-encoder needs (§4.2).

**Table 4: Hits@1 with 95% bootstrap CIs for the 2×2 cross-encoder train/evaluation matrix; row labels name the training miner. The zero-shot E5 row is bootstrapped separately (within 0.002 of Table 1). The advantage over E5 holds in headline pairings (+20 to +40 pp) and swapped (+13 to +39 pp); re-trained `trn-E5` cells reproduce the original within 2 pp on time/event.**

| | Eval on benchmark pool (`pairs.parquet`) | | | Eval on TF-IDF pool (`pairs_v3.parquet`) | | |
|---|---|---|---|---|---|---|
| | TIME | EVENT | JURIS | TIME | EVENT | JURIS |
| E5-small zero-shot | 0.204 ±1.8 | 0.232 ±2.1 | 0.432 ±10.5 | 0.265 ±1.4 | 0.320 ±1.7 | 0.441 ±7.8 |
| cross-enc trn-E5 | **0.601** ±2.2 | **0.552** ±2.4 | 0.705 ±8.8 | 0.544 ±1.8 | 0.529 ±1.7 | 0.571 ±7.5 |
| cross-enc trn-TF-IDF | 0.589 ±2.0 | 0.534 ±2.4 | **0.750** ±9.1 | **0.655** ±1.5 | **0.615** ±1.7 | **0.640** ±7.1 |
| Δ *vs. E5, headline pairing* | +39.7 | +32.0 | +27.3 | +39.0 | +29.5 | +19.9 |
| Δ *vs. E5, swapped pairing* | +38.5 | +30.1 | +31.8 | +27.9 | +20.9 | +13.0 |

## 5.8 Additional diagnostics

Four diagnostics; full tables ship with the released code. **B-cubed cluster F1** [2] is 0.92 / 0.90 / 0.91 for the name+state cross-encoder and 0.94 / 0.93 / 0.94 for the full-record one, vs. ≤0.48 for any bi-encoder (each collapses to one giant cluster at the tuned threshold); restricted to true clusters of size ≥2, 0.85 / 0.84 / 0.88 (name+state). **Calibration and selective prediction.** ECE (10-bin) is 0.20 and Brier 0.18 for the name+state cross-encoder (0.13 / 0.12 full-record); temperature scaling does not fix the high-score over-confidence, so we report raw scores. A selective-prediction [5] sweep on the ensemble reaches Hits@1 0.93–0.98 at 20% coverage and 0.83–0.98 at 50%, so high-precision users can threshold. **Cross-domain transfer.** Zero-shot Magellan Walmart-Amazon [6]: the cross-encoder *loses* to its own zero-shot base encoder by 5.9 pp AUROC. The fine-tuned representation is specialized to lifecycle-spanning legal-entity matching, not generic ER. **Label quality, second opinion.** A rule-based re-classifier, orthogonal to the primary's event-type evidence, agrees with it on 98.7% [95% CI 97.0–100%] of the 231/277 confidently-judged positives (46 uncertain), bounding detectable label noise there at ≈3%; human review is future work.

## 5.9 Where each method is fooled

Table 5 reports the mean score per negative class, where higher means more often fooled. Bi-encoders cut zero-shot E5's score on `parent_subsidiary` and `same_name_diff_country` by 40–52%, but `same_name_same_country` remains a wall at ≈0.89. The cross-encoder breaks it (0.89 → 0.43; 0.36 ensemble), and the ensemble nearly solves the other name classes (0.91 → 0.17 / 0.28). The one regression is `parent_subsidiary` (0.50 → 0.67 → 0.75 across bi-encoder, L-6, ensemble): the prompt enrichment that injects GLEIF L2 parent/child names makes it *worse*. At 99 held-out examples the headline impact is small, but the direction is clear: explicit multi-class modeling of corporate-hierarchy relationships.

**Table 5: Negative-class mean score (lower is better) on negatives *never seen in training* (*n*=99 / 1,398 / 1,096 / 20,759): *p/s* parent/subsidiary, *sn-D* same-name/diff-country, *sn-S* same-name/same-country, *tok* token-overlap. The ensemble improves three classes but *worsens p/s* (0.67 → 0.75): prompt-injected L2 context makes parent/subsidiary pairs look *more* similar. Scores are on each method's own scale (compare down columns); bold = lowest among learned methods.**

| Method | p/s | sn-D | sn-S | tok |
|---|---|---|---|---|
| `tfidf_char` | 0.220 | 0.590 | 0.783 | 0.595 |
| `sentence_encoder` | 0.841 | 0.912 | 0.960 | 0.905 |
| `method_full` | 0.501 | 0.438 | 0.893 | 0.747 |
| Cross-encoder (L-6, 22M) | 0.668 | 0.201 | 0.432 | 0.311 |
| **Ensemble (L-6×L-12)** | 0.748 | **0.166** | **0.358** | **0.284** |

# 6 ANALYSIS

## 6.1 Error analysis

We sampled 100 `method_full` failures on held-out time: 66 mixed or other, 19 where anchor and true successor share *no* tokens, 8 substring-overlap distractors, and 7 exact normalized-name collisions. The no-token cases are hardest for name-input methods — they need world knowledge, graph context, or the full record (§5.4); Figure 1's anchor is representative: resolving "DB" to Deutsche Bank from name strings alone is beyond what a 33M encoder reliably does.

*Residual JURIS failures cluster on sibling consolidation.* Of the 19 / 88 anchors the ensemble mis-ranks on JURIS, at least 11 are sibling consolidations — the model returns a sibling vessel instead of the parent — plus Japanese-script renames the ASCII tokenizer cannot bridge. Multilingual pretraining and ownership-graph context would address both.

# 7 LIMITATIONS AND FUTURE WORK

*The "anti-correlation" finding is a construction artifact.* It follows from mining name-similar negatives against name-different positives (§5.4), and motivates joint encoding here, not a universal claim about name similarity in ER.

*In-distribution headline overstates deployment.* The headline Hits@1 is measured on the released pool; end-to-end on the fielded pipeline every reranker falls to 15–39% (§5.6) because bi-encoder Recall@20 is 33–50%, so the reranker often never sees the true successor. A stronger first stage (hybrid sparse–dense, or a pre-filtered shortlist) is the highest-leverage future work.

*Seed variance and a sliver of mislabeled hard negatives.* Over seeds 42/7/1337 the bi-encoder varies ±0.5/0.4/1.6 pp and the enriched cross-encoder ±0.5/0.3/1.6 pp (TIME/EVENT/JURIS). The new headline systems are seed-stable: on-pool the full-record GBM moves ≤0.5 pp and the cross-encoder 1.2–3.4 pp; fielded-conditional, both stay within ≈4 pp on TIME/EVENT (wider on 80-anchor JURIS); there the cross-encoder's *worst* seed still leads the GBM's best by 8 pp, narrowing to 2.5 on JURIS. Separately, 205/93,695 = 0.22% of hard negatives appear in GLEIF's `SuccessorEntity`; dropping them moves results ≤0.2 pp.

*Survivorship bias and thin slices.* GLEIF `SuccessorEntity` is populated voluntarily, which biases positives toward well-counseled regulated entities; of 302 marquee Wikidata M&A pairs, 283 are absent from it (§3.4), so high-profile events are under-represented. The held-out-event test set is 94.5% absorption, and the jurisdiction test has only 88 eligible anchors (±8.5 pp CI). Gains clear the CI band on all three splits, but the per-split absolutes should be read with those sample sizes in mind. Eligibility itself conditions on a name-colliding neighbour existing, which amplifies the construction (§5.2); a 277-pair second-classifier check bounds detectable label noise at ≈3% (§5.8). Scored against *random* same-country pools rather than mined ones, the marquee supplement (283 pairs, 244 anchors) inverts the picture. Token Jaccard reaches 0.41, the name+state cross-encoder falls below chance (0.07), and the full-record models recover only partly (0.21–0.24). A model trained on mined pools inherits their anti-similarity prior, and should not be assumed to transfer to unmined candidate distributions. Finally, splits are pair-level rather than entity-disjoint (32% / 21% of time-split test candidates / gold successors appear in training pairs; JURIS <1%), though the L-6 shows no seen-entity advantage (unseen 0.62 vs. seen 0.57 on TIME, $p$=0.04 toward unseen; 0.57 vs. 0.58 on EVENT, n.s.); disjoint splits are a stricter future design.

*LLM baselines: what the discrete scale hides.* A 0–10 rubric caps what ranking can express: on TIME/EVENT the five API models forfeit 13–48 pp Hits@1 to within-pool ties. Gemini 3.1 Pro scores every candidate identically on 55% of TIME anchors, so oracle tie-breaking would lift it to 0.72 from the 0.23 we measure — above our L-6 — whereas Sonnet and Opus collapse only ≈5% of pools, and the cross-encoder's continuous scores pay a ≤0.6 pp tie tax. Prompt format moves results too: batched Sonnet 4.6 scores 0.46 on TIME against 0.39 single-pair on the same 56 anchors. The panel therefore bounds what this elicitation format extracts, not what these models can do. Finer scales, log-probs, or comparative tie-breaks would likely raise them, as would a full-record prompt (the LLM analogue of §5.4's finding), and cross-vendor gaps may partly reflect exposure to GLEIF-derived text.

*The model is specialized, and parent/subsidiary is a thin slice.* The cross-encoder underperforms zero-shot E5-small on out-of-domain product matching (Walmart-Amazon, −5.9 pp AUROC; §5.8): the representation is specialized to lifecycle ER, not generic. The full-record cross-encoder closes the input gap (§5.4) but pays the usual joint-encoding inference premium over trees; a full-record *LLM* prompt is the remaining untested cell. Parent/subsidiary negatives regress (0.50 → 0.67 → 0.75 across bi-encoder, L-6, and ensemble), though at 99 held-out examples the effect is weak; explicit multi-class modeling of related-but-distinct entities is clear future work.

## 8 CONCLUSION

Sylvester is an open benchmark for lifecycle-aware entity resolution on financial entities, built entirely from public GLEIF, SEC EDGAR, and Wikidata data around a three-pool protocol that separates methods which learn the pool from methods that learn the task. Both sides of it are the story. Name-only exploits post the paper's best name-input scores and then collapse off-pool; among name-input methods, a single 22M-parameter cross-encoder survives, staying ahead of every zero-shot LLM we test on the two large splits. But input matters as much as architecture: boosted trees over the full structured record transfer across all three pools at negligible cost, and a cross-encoder over that same record holds onto the most signal under shift, winning the fielded universe by 12 pp. Much of the lifecycle signal, in the end, is in the record. What remains open is the first stage: even the best reranker is capped by 33–50% retrieval recall, so in our fielded simulation it is candidate generation, not matching, where lifecycle ER breaks.

*Reproducibility.* Code (MIT), data (CC-BY-4.0), parsers, predictions, trained weights, and a datasheet: https://github.com/chang-07/er-research; archival DOI on acceptance. Every non-LLM result reproduces from raw GLEIF in ≈2 hours on one MacBook.

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